Carnegie Mellon University

"ACME" Motors

Final Capstone Presentation

Tiffany, Beihua, Kerri, Katherine

Our Team

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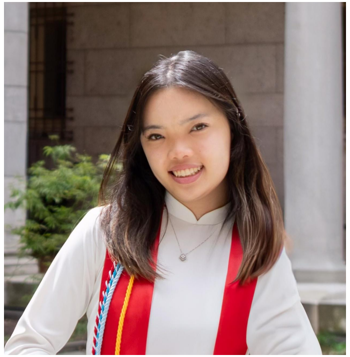
Our team



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Our Team

Tiffany T. Nguyen

Background:

Business, Finance, Data Science Undergrad

Goldman Sachs Client Engagement Co-Op

Scotiabank Markets Co-Op

Capco Consulting Intern

Katherine Cornell

Background:

Business Undergrad, Indiana University

Compensation Analytics, PNC Bank -- 3 yrs

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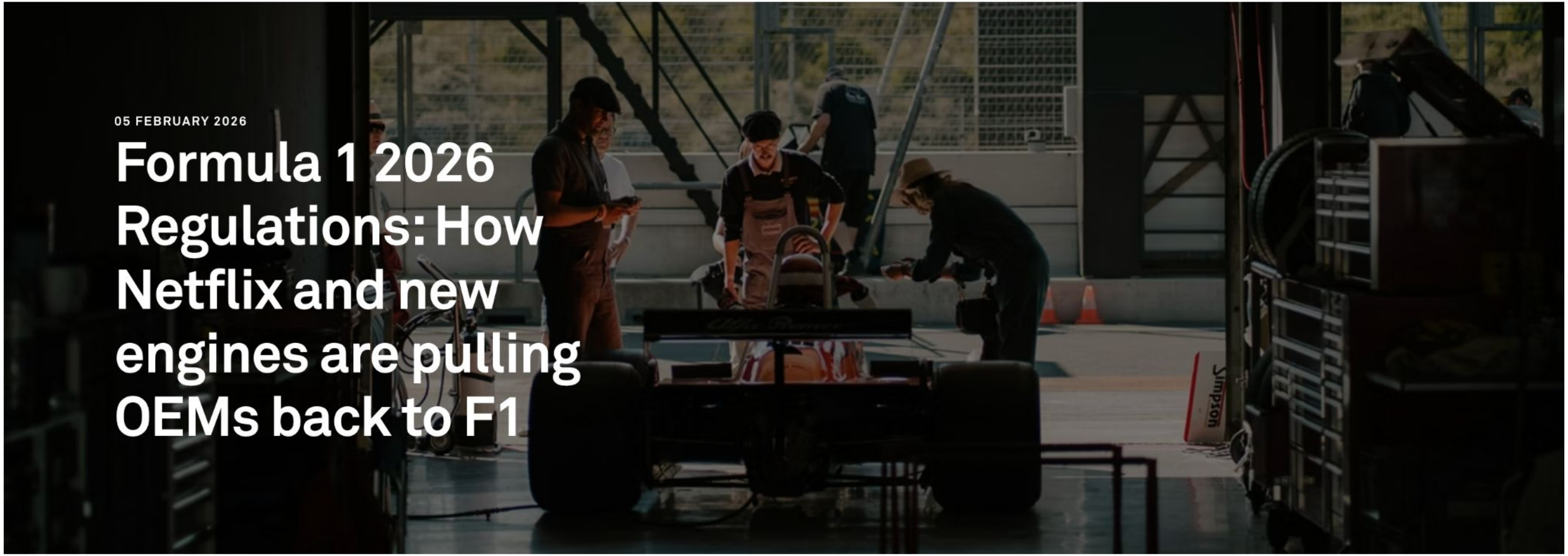

05 FEBRUARY 2026

Formula 1 2026 Regulations: How Netflix and new engines are pulling OEMs back to F1

ADVERTISING

Tech, AI players help drive F1 sponsorship spend past $3B in 2026

By Bevin Fletcher · Feb 23, 2026 11:28am

live sports | sports streaming | sponsorships | Apple TV

Industry Relevance:

With increasing sports racing industry and consumer interest climbing...

How can companies leverage **data analytics** into their
car feasibility modeling?

Industry problems

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Industry problems

1. Motorsport entry decisions involve high uncertainty and capital risk

Multi-millions investment	Execution risk
Performance balancing constraint	Market Changes / Trends

2. Decision Challenges (Complexity)

Complex Tiers into Decision	Fluctuations in Demand
Conversion Costs	Comp. OEM Landscape

Industry problems

Motorsport entry decisions involve high uncertainty and capital risk

- GT racing programs require multi-million dollar development investment
- Homologation timelines and technical compliance create execution risk
- Performance balancing (BoP) introduces competitive uncertainty
- OEMs must commit before market demand is fully known

Decision Challenge

- OEMs face a complex go / no-go decision
- Multiple GT tiers with different requirements
- Different conversion costs by platform
- Uncertain customer demand
- Competitive OEM landscape

Why this matters？

GT racing can generate significant strategic value

- Brand halo and performance credibility
- Customer racing revenue (cars + spares + support)
- Dealer and enthusiast activation
- Platform for tiered racing ladder

Why this matters？

GT racing can generate significant strategic value

Revenue	Brand Association	Enthusiast Activation
Customer **racing revenue** (cars + spares + support)	Brand **halo** and **performance credibility**	Dealer and **enthusiast activation**

Data analysis

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Dataset Structure

We use a combination of structured public proxies and synthetic datasets organized to support the stage-gate framework.

Strategy & Brand

- Brand KPI weights (A_brand_kpis)
- Competitive activation benchmarks (B_competitive_benchmark)

Customer & Market

- Customer personas and preferences (C_personas)
- Regional team demand proxies (I_team_tam)

Engineering & Feasibility

- Production platform inventory (D_platform_inventory)
- Regulatory fit and conversion burden (E_reg_fit)
- Conversion BOM proxy (Synthetic_BOM_Proxy)

Dataset Structure

We use a combination of structured public proxies and synthetic datasets organized to support the stage-gate framework.

Competitive & Risk

- BoP volatility risk index (F_bop_risk)
- GT series constraint assumptions (GT_Series_Assumptions)
- Competitor economics benchmarks (J_competitor_economics)

Financial Modeling

- Unit economics model (Unit_Economics_Model)
- Stage-gate decision log (O_stage_gate_log)

Data insights

- Evaluate **entry** across strategy, engineering, market, and finance
- Define **feasibility, conversion burden, and homologation risk** through platform and regulatory data
- Model demand, pricing, and launch markets via proxies & personas
- Benchmark competitors on cost, positioning, and volatility
- **Output: Go // Conditional Go // No-Go Decisions**

Modeling process

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Model 1

Overarching Question

Given synthetic raw car data...

What model can calculate the best fit car for certain racing circuits?
What gateway inputs / outputs drive the model?

Model 1

Input: synthetic data across strategy, engineering, market, and economics

Gate 0: brand intent scoring

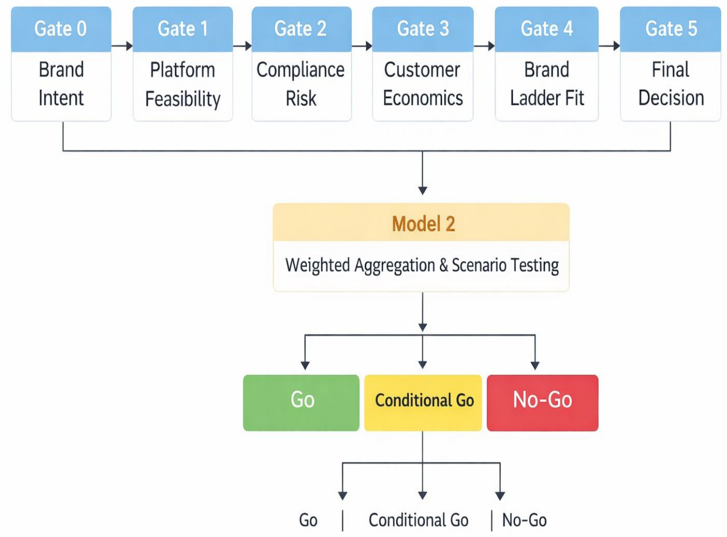
Gate 1: platform feasibility filtering

Gate 2: compliance and BoP risk scoring

Gate 3: customer economics evaluation

Gate 4: brand ladder fit scoring

Output: platform–tier level recommendation with gate-level scores



Gate 0 (Brand Intent)

Inputs:

Brand KPIs, Baseline, Year 1 Target, Weight, Pillar

Competitor earned media, social index, racial emphasis

Lift = (target year - baseline) / baseline

Contributor = lift * weight

Brand intent score = (sum all contr) * 100

Pillar weights = grouping by brand pillar

Outputs: Brand intent score baseline & competitor benchmarks

Hard filters and thresholds

Gate 1 (Platform Feasibility)

Inputs:

Platform inventory
(avail, heritage, stability score)

Feasibility flag, BOM cost Proxy, Compliance Risk Score

Remove if feasibility flag = NO

Remove Sunset // Compliance Risk > 0.75

Platform score = avg global avail, heritage, platform stability score

Feasibility score = (avg platform score / 5) * 10 * (1- compliance risk)

Outputs: Shortlist of cars that cleared “kill” conditions

Gate 2 (Compliance Risk)

Inputs:

BOP volatility index, drivability score, compliance risk, Gate 1 short list

Balance of Perf. Data
Regulatory Fit Data

Combined risk = BOP volatility * 5 + (compliance risk * 5)

Risk flag = BOP vol. > 0.55

Best candidates with combined risk

Outputs: Gate 1 Short lists with risk scores attached

ROI protected

Gate 3 (Customer Economics)

Inputs:

Active cars, new buys per season, race cars, MSRP, annual budget, gross margin

Economics Raw Data
Personas Raw Data

TAM Total = sum active cars ; TAM New Buys = sum new purchases

Projected 3 year units = (new buys) * 10% growth rate

Econ score = 10 if projected > = breakeven
(projected / breakeven) * 10 otherwise

Annual profit = (gross margin * race msrp) + annual spares rev - annual support cost

Outputs: *Shortlist with econ score + annual profit estimate*

Hard filters and thresholds

Gate 4 (Brand Ladder Fit)

Inputs:

Annual budget estimates
Earned Media Index
Brand KPIs (from prior gates)

Peer Raw Data
Personas Raw Data

Ladder coverage = share of personas with GT4 and GT3 budget

Avg activation = mean (earned media + social index) / 2

Ladder score = (ladder_coverage * 4) + (Avg activation / 100 * 3) + (kpi_alignment * 3)

(Mult. by Weights)

Outputs: *Shortlist with ladder score*

Gate 5 (Final Weighted Score)

Risk score = (10 - combined risk)

Total score = (brand score * 0.20)
+ (plat_score * 0.20)
+ (risk_score * 0.20)
+ (econ_score * 0.25)
+ (ladder_score * 0.15)

OUTPUT: gate 5 – ranked candidates

nameplate	target_class	gt_series_tier_recommendation	plat_score	risk_score	econ_score	total_score
Ford Mustang (S650)	GT3	Advanced Tier	5.333333	8.05	2.04	6.42
BMW M4	GT4	Entry Tier	5.266667	7.75	1.45	6.20
Mercedes-AMG GT	GT4	Entry Tier	3.786667	5.25	3.09	5.82
Toyota GR Supra	GT3	Advanced Tier	2.560000	6.30	1.73	5.44
BMW M4	GT3	Advanced Tier	4.666667	4.60	1.37	5.43
Mercedes-AMG GT	GT3	Advanced Tier	2.400000	6.35	1.33	5.32
Porsche 718 Cayman	GT3	Advanced Tier	1.680000	4.40	1.17	4.75
Chevrolet Corvette (C8)	GT3	Advanced Tier	1.040000	3.60	2.28	4.74

Outputs: *Winners & Candidates with total score descending*

Model 2

Reverse Analysis:

Given specific **car model (X)** and **specific desired racing circuit (Y)**...

what is the **cost estimate** to enable racing feasibility?

Model 2: Breakdown Analysis

1. Pull **base car specifications** (synthetic data set)

2. Check car's **regulatory fit** (depending on GT class)

3. Estimate car conversion materials bill (BOM)

4. Note risk flags (risk score / assembly complexity)

5. Compute rule-based feasibility score

   - Regulatory score
   - Conversion cost score
   - Compliance risk score
   - Complexity score
   - Platform stability score

6. Decision (Go / No-Go) based on analysis

Model 3

Logistic Regression

Given the existing synthetic data…

What are the ***drivers (top factors & coeff)*** that contribute to the **final go / no decision?**

```
Top factors pushing toward GO (positive):
--------------------------------------------------
  platform_stability              +0.623
  cost_ratio                      +0.569
  curb_weight_lb                  +0.548
  breakeven_units                 +0.279
  motorsport_heritage             +0.235
```

Model 3

Top Factors Pushing towards GO decision (positive)

Platform Stability	Cost Ratio	Curb Weight (pounds)
Strongest signal (strong OEM platform to start)	**Targeted upgrades** often require investment Cars with low cost ratio scored low in other categories)	Heavier cars tended to be **more premium** (proxy for platform tier and brand seriousness)

Model 3

```
Top factors pushing toward NO-GO (negative):
------------------------------------------------------------
  avg_serviceability           -0.346
  annual_spares_rev            -0.346
  avg_assembly_complexity      -0.366
  eng_complexity               -0.775
  compliance_risk              -1.158
```

Top Factors Pushing towards NO GO decision (negative)

Avg Serviceability

Higher servability =
Difficult future maintenance

Avg Spares Revenue

Higher spare rev = signal that car may break down faster

Assembly Complexity

Higher assembly complexity = expensive car production / timing

Unique classes: [0 1] – shape: (16, 20)

LR accuracy (cross-val): 0.622 ± 0.031

Model 3 Improvements

Logistic Regression on all features

Focusing on targeting features

Reduces risk of overfitting

Improving Baseline

62.2 accuracy (cross-val) with full features

Less features = more model improvement projected by 10-15%

Questions We Answered

Go / No-Go Model: How It Works

Dimension	Weight	Gate	How It's Calculated
Brand Intent	**20%**	0	Weighted KPI lift × brand pillar weights, capped at 10
Platform Fit	**20%**	1	Mean of availability, heritage, stability scores × (1 − compliance risk)
Risk	**20%**	2	10 − combined risk (BoP volatility × 5 + compliance risk × 5)
Economics	**25%**	3	Projected 3yr units / breakeven units × 10 at 10% capture rate
Brand Ladder	**15%**	4	Ladder coverage × 4 + activation × 3 + KPI alignment × 3

- **Go: Total Score ≥ 6.0**
- **Conditional Go: 5.0 ≤ Total Score < 6.0**
- **No-Go: Total Score < 5.0**

Final Candidate Ranking (Gate 5 Output)

Rank	Vehicle	Class	Platform	Risk	Econ	Total	Decision
1	Ford Mustang (S650)	GT3	5.33	8.05	2.04	**6.42**	**Go**
2	BMW M4	GT4	5.27	7.75	1.45	**6.20**	**Go**
3	Mercedes-AMG GT	GT4	3.79	5.25	3.09	**5.82**	Cond. Go
4	Toyota GR Supra	GT3	2.56	6.30	1.73	**5.44**	Cond. Go
5	BMW M4	GT3	4.67	4.60	1.37	**5.43**	Cond. Go
6	Mercedes-AMG GT	GT3	2.40	6.35	1.33	**5.32**	Cond. Go
7	Porsche 718 Cayman	GT3	1.68	4.40	1.17	**4.75**	No-Go
8	Corvette (C8)	GT3	1.04	3.60	2.28	**4.74**	No-Go

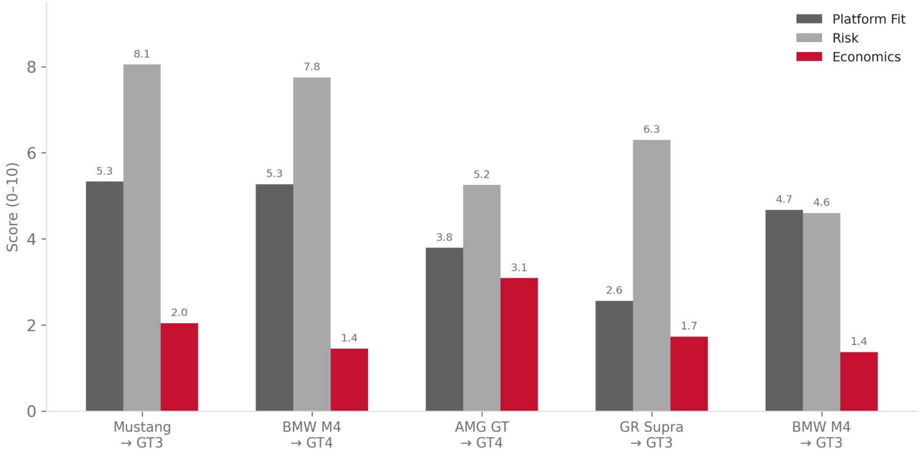
Brand Score and Ladder Score are constant across all candidates (applied uniformly from Gates 0 and 4), so variation in ranking is driven entirely by Platform Fit, Risk, and Economics.

Top 5 Candidates by Total Score

Callout 1: Ford Mustang (S650) → GT3 scores 6.42. Highest platform fit (5.33) among all candidates and lowest combined risk, yielding the top rank.

Callout 2: Only 2 of 8 candidates cross the Go threshold (≥ 6.0). The remaining 3 in the top 5 fall into Conditional Go territory (5.43 to 5.82).

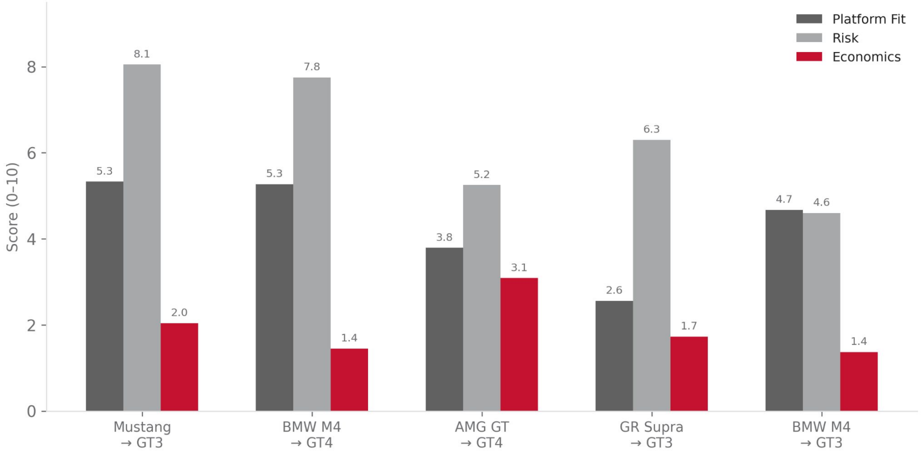
Where Each Candidate Gains and Loses Points



1. Risk scores are the **highest** dimension across top candidates:

Most pairings carry manageable BoP and compliance risk

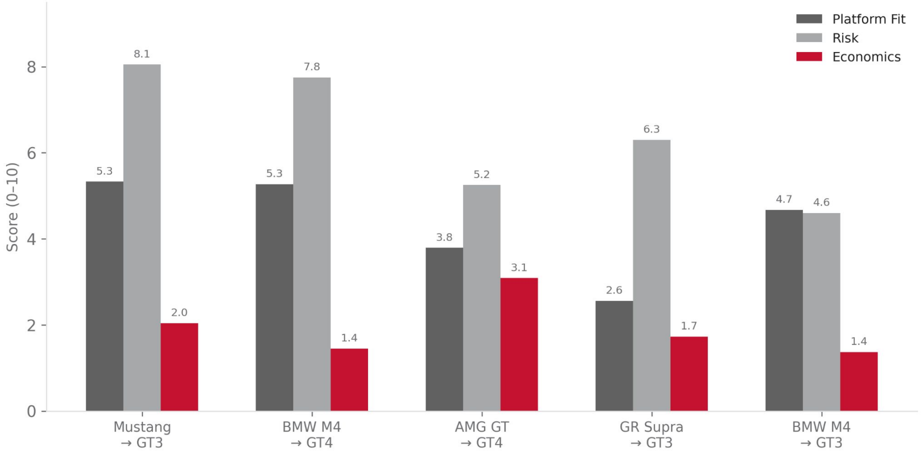
Where Each Candidate Gains and Loses Points



2. Econ scores are the **lowest** across all 8 candidates (max 3.09):

Driven by the 10% capture rate assumption and high breakeven thresholds

Where Each Candidate Gains and Loses Points

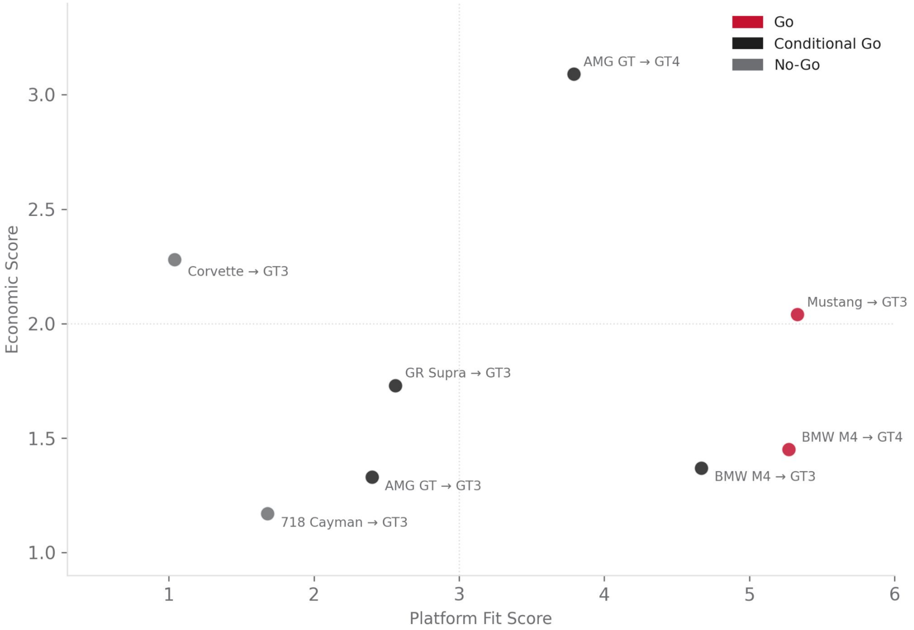


3. Platform fit is the **differentiator**:

Top 2 candidates score above 5.0

Mid-tier candidates fall below 4.0

Platform Fit vs Economic Viability



No candidate scores high on both axes: entry decisions are tradeoffs.

Corvette: low fit, moderate econ. Platform weakness is not offset.

Mustang/BMW M4: best fit, but econ still below 2.1.

AMG GT → GT4: best econ (3.09), but fit keeps it Conditional.

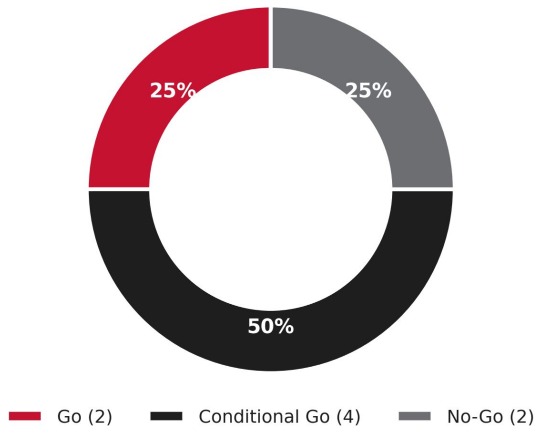
Results

High-Level Results

2 / 8
Go

4 / 8
Conditional Go

2 / 8
No-Go



1. **Technical readiness ≠ viability**. Mustang: highest fit (5.33), econ only 2.04.

2. Conditional Go candidates **fail on one dimension** only, meaning targeted fixes can shift them.

3. Economics is the **binding** constraint: avg econ score across all 8 is 1.80 / 10.

4. The framework is **diagnostic**: it identifies which dimension holds a candidate back.

Risks & Limitations

- **Synthetic data:** all inputs constructed, not from real OEM/FIA records.
- Brand + Ladder scores are **uniform** across all candidates (35% of weight does not differentiate).
- Fixed **10% capture rate** for all candidates.
- No sensitivity analysis on weight configuration.
- Scope limited to **GT3/GT4** only.

Forward-Thinking Implementation

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What's Next

Model Iterations:

1. Real FIA/SRO homologation data
2. Sensitivity analysis on weights
3. Candidate-specific brand + ladder scores

Generic Use:

1. Any OEM plugs in their platform + target class
2. Reusable for portfolio reviews and sponsor pitches

Beyond Racing:

1. Any high-investment product entry decision
2. EV variants, special editions, fleet programs

Questions

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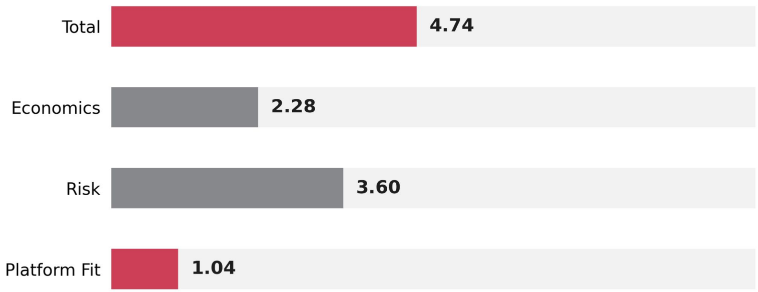
Appendix

Case Study: Chevrolet Corvette (C8) → GT3



Recommendation: No-Go (Total Score: 4.74, Rank 8 of 8)

Platform Fit: 1.04 — Lowest of all 8. Low availability/heritage/stability scores compounded by high compliance risk.

Risk: 3.60 — Combined risk = 6.40 (high BoP volatility + compliance risk). Score = 10 − 6.40.

Economics: 2.28 — 3-year projected units at 10% capture fall short of breakeven.